

National University of Computer and Emerging Sciences
Islamabad Campus

Statistical Modeling (MT2002)

Course Instructor(s):

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Section(s): A & B & C

Final Examination

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 6

Date: Jun 10, 2025

Roll No

Course Section

Student Signature

Do not write below this line.

1. Solve all questions.
2. Clearly mention the question number as well as the part number of each question.
3. The source code should be properly indented.
4. Simply providing the answer is not enough to get the marks. Justify your answers.
5. Solve all questions and all parts of each question in order to get a bonus of 2 marks.

Q1:

[10 marks]

Consider a biased coin with the probability of heads (1's) higher than the tails (0's). Write the pymc code, using the inference button, for the coin flipping problem using Bayesian inference.

Q2:

[5+5 marks]

- A) Explain why normal (Gaussian) distribution is not a good choice when it comes to inference using the data with outliers?
- B) What would be a better option in that case and why? Explain with an example.

Q3:

[5+5+5 marks]

Provide brief answers:

- A) What is highest posterior density (HPD)?
- B) What is the region of practical equivalence (ROPE)? Explain with an example.
- C) Once the ROPE is defined we compare it with HPD the following three scenarios arise. Tell whether the coin would be fair in these scenarios.
 - a. The ROPE contains the entire HPD
 - b. The ROPE does not overlap with the HPD
 - c. The ROPE partially overlaps with the HPD

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Q4:

[8 marks]

You are analyzing admission decisions for a university program. You are given a dataset containing binary outcomes indicating whether a student was admitted (1) or not (0), based on two features:

1. hours_studied: the number of hours a student studied for the entrance exam.
2. GPA: the student's grade point average.

You want to model the probability of admission using Bayesian logistic regression with PyMC. Write a complete PyMC model that does the following:

1. Uses two features: hours_studied and GPA to predict admitted.
2. Defines appropriate priors for the intercept and coefficients.
3. Applies a logistic transformation to compute the probability of admission.
4. Defines the likelihood using a Bernoulli distribution.
5. Defines the decision boundary (as we might be interested in visualizing the uncertainty in the decision boundary).
6. Samples from the posterior using NUTS.
7. Visualizes the posterior distributions of the model parameters using ArviZ.

Solution

```
import numpy as np
import pandas as pd
import pymc as pm
import arviz as az
import matplotlib.pyplot as plt
import seaborn as sns

# Simulate example data
np.random.seed(42)
n = 100
hours = np.random.normal(5, 2, n)
```

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```
gpa = np.random.normal(3, 0.5, n)
logits_true = -4 + 0.8 * hours + 1.5 * gpa
prob = 1 / (1 + np.exp(-logits_true))
admitted = np.random.binomial(1, prob)
data = pd.DataFrame({"hours": hours, "gpa": gpa, "admitted": admitted})

# Bayesian logistic regression model
with pm.Model() as model:
     $\alpha$  = pm.Normal(" $\alpha$ ", 0, 10)
     $\beta_{\text{hours}}$  = pm.Normal(" $\beta_{\text{hours}}$ ", 0, 10)
     $\beta_{\text{gpa}}$  = pm.Normal(" $\beta_{\text{gpa}}$ ", 0, 10)

    logits =  $\alpha$  +  $\beta_{\text{hours}}$  * data["hours"] +  $\beta_{\text{gpa}}$  * data["gpa"]
     $\theta$  = pm.Deterministic(" $\theta$ ", pm.math.sigmoid(logits))

    y_obs = pm.Bernoulli("y_obs", p= $\theta$ , observed=data["admitted"])

    x_vals = np.linspace(data["hours"].min(), data["hours"].max(), 100)
    decision_boundary = pm.Deterministic("decision_boundary", -( $\alpha$  +  $\beta_{\text{hours}}$  * x_vals) /  $\beta_{\text{gpa}}$ )

    trace = pm.sample(2000, tune=1000, target_accept=0.9, return_inferencedata=True)

# Posterior distribution plots
az.plot_posterior(trace, var_names=[" $\alpha$ ", " $\beta_{\text{hours}}$ ", " $\beta_{\text{gpa}}$ "], kind="hist", credible_interval=0.95)
plt.tight_layout()
plt.show()
```

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Q5:

[4+4+4 marks]

The following table shows the results of comparison of three models with ArviZ.

	waic	waic_se	p_waic	loo	loo_se	p_loo	scale
Model_1	9.13	5.35	2.67	8.92	5.24	2.56	deviance
Model_2	28.77	4.62	2.46	28.72	4.53	2.45	deviance
Model_3	8.77	8.62	2.96	8.72	6.53	2.86	deviance

Now answer the following questions (each question not more than 3 lines):

- A) Which model is the simplest of all? Justify your answer by referring to the appropriate columns.
- B) Which model most accurate of all? Justify your answer by referring to the appropriate columns.
- C) Which model is the best of all? Justify your answer by referring to the appropriate column.

Solution

A) Which model is the simplest of all?

Model_2. It has the lowest p_waic (2.46) and lowest p_loo (2.45), indicating the smallest effective number of parameters and hence the least complexity.

B) Which model is most accurate of all?

Model_3. It has the lowest WAIC (8.77) and lowest LOO (8.72), both of which indicate better predictive performance on the deviance scale (lower is better).

C) Which model is the best of all?

Model_3. It combines low complexity (p_loo = 2.86) with the highest accuracy (lowest LOO = 8.72), making it a good trade-off between simplicity and fit.

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Q6:

[1 x 45 = 45 marks]

Select the correct option for each of the following questions. Fill the attached bubble sheet to get marks. Multiple marking, overwriting, unclear marking will result in zero marks in that part.

1. Which model is most appropriate to predict a continuous outcome from a single continuous predictor using PyMC?

- A) Generalized linear model
- B) Simple Linear model
- C) Softmax Regression
- D) None of a,b,c

2. Which model should you choose when you want to predict a binary outcome using multiple features?

- A) Simple Linear Model
- B) Multiple Logistic Regression
- C) Softmax Regression
- D) Bayesian Linear Regression

3. If your outcome (dependent variable) has more than two categories that don't follow any order, which model should you use?

- A) Bayesian Logistic Regression
- B) Linear Regression
- C) Softmax Regression
- D) Polynomial Regression

4. A non-linear relationship between an independent and dependent variable is observed. What should you use?

- A) Hierarchical Linear Model
- B) Softmax Regression
- C) Polynomial Regression
- D) Logistic Regression

5. You want to model group-level variation in slopes and intercepts. Which model suits

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best?

- A) Bayesian Linear Regression
- B) Polynomial Regression
- C) Hierarchical Linear Model
- D) Logistic Regression

6. When adding new variables to a multiple linear model causes the effect of some predictors to appear weaker or disappear, this phenomenon is known as:

- A) Class imbalance
- B) Overfitting
- C) Masking effect
- D) None of A,B,C

7. Which issue in multiple regression arises when two predictors are highly correlated?

- A) Confounding
- B) Masking
- C) Multicollinearity
- D) Class imbalance

8. Which model is most appropriate when dependent variable is count variable?

- A) Logistic Regression
- B) Softmax Regression
- C) Hierarchical Model
- D) Poisson Regression

9. Which model is appropriate when your count data has many zeros, and you want to model both the excess zeros and the count outcomes separately?

- A) Simple Linear Model
- B) Zero-Inflated Poisson (ZIP) Model
- C) Logistic Regression
- D) Softmax Regression

10. Which model best handles varying intercepts and slopes across grouped observations?

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- A) Multiple Linear Regression
- B) Hierarchical Linear Model
- C) Softmax Model
- D) Logistic Regression

11. When some variables influence both the predictors and the outcome (confounders), what is the best way to handle them in your statistical model?

- A) Increase the sample size
- B) A & D
- C) Include the confounding variables in the model to adjust their effects
- D) Use logistic regression regardless of the confounders

12. What modeling choice helps address multicollinearity?

- A) Use class balancing
- B) B) Use uncorrelated predictors
- C) A & B
- D) None of A,B,C

13. Which model is best for binary classification with a single input feature?

- A) Softmax Regression
- B) Hierarchical Linear Model
- C) Generalized linear model
- D) Polynomial Regression

14. Which modeling approach should be preferred if response distribution differs by group?

- A) Softmax Regression
- B) Hierarchical Model
- C) Logistic Regression
- D) Simple Linear Model

15. In a Zero-Inflated Poisson (ZIP) model, what do the parameters ψ and θ mean?

- A) ψ is the chance of extra zeros, and θ is the average count of events
- B) ψ is the average count, and θ is the chance of extra zeros

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C) only A & B

D) ψ is the slope, and θ is the intercept

16. What is the main purpose of posterior predictive checks in Bayesian modeling?

A) To estimate model parameters

B) To assess model fit by comparing observed and predicted data

C) To calculate Bayes factors

D) None of A, B, C

17. Which visualization is commonly used in posterior predictive checks to compare the distribution of observed data with replicated data?

A) Trace plots of parameter samples

B) KDE (Kernel Density Estimate) plots of observed vs. predicted data with p-values

C) Confusion matrices

D) None of A, B, C

18. What does WAIC stand for?

A) Weighted Average Information Criterion

B) Widely Applicable Information Criterion

C) Weighted Approximate Importance Calculation

D) None of A, B, C

19. Both WAIC and LOO are examples of:

A) Parameter estimation methods

B) Predictive accuracy metrics

C) Sampling algorithms

D) None of A, B, C

20. In LOO cross-validation, what does LOO stand for?

A) Logarithm of Observations

B) Leave-One-Out

C) Likelihood of Observations

D) None of A, B, C

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21. Information criteria values reported on the negative log scale are typically:

- A) Minimized for better models
- B) Maximized for better models
- C) Irrelevant for model comparison
- D) None of A, B, C

22. A lower WAIC value with log scale indicates:

- A) Better model predictive performance
- B) Poorer model fit
- C) Larger prediction error
- D) None of A, B, C

23. What is the main use of Bayes factors in model comparison?

- A) To calculate posterior predictive checks
- B) To quantify evidence in favor of one model over another
- C) To estimate model parameters
- D) None of A, B, C

24. Bayes factors compare:

- A) Prior distributions only
- B) Marginal likelihoods of competing models
- C) Posterior means
- D) None of A, B, C

25. A Bayes factor greater than 1 indicates between model 1 and model 2:

- A) model 1 is better
- B) model 2 is better
- C) No difference between models
- D) None of A, B, C

26. One limitation of Bayes factors is:

- A) They require large sample sizes only

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- B) Sensitivity to the choice of prior distributions
- C) They cannot compare nested models
- D) None of A, B, C

27. What is the primary goal of multiple linear regression?

- A) To model the relationship between one dependent variable and multiple independent variables
- B) To classify data into discrete categories
- C) To find clusters in data
- D) To minimize classification error

28. Which of the following statements about Bayesian multiple linear regression is true?

- A) It treats model parameters as random variables with probability distributions.
- B) It assumes fixed coefficients, similar to Ordinary Least Squares Method.
- C) It does not allow for uncertainty estimation.
- D) It cannot be implemented using PyMC.

29. In PyMC, which distribution is commonly used as a prior for regression coefficients in Bayesian multiple linear regression?

- A) Normal
- B) Uniform
- C) Exponential
- D) Poisson

30. Which sampling method is often used in PyMC to estimate posterior distributions in Bayesian regression models?

- A) Markov Chain Monte Carlo (MCMC)
- B) Principal Component Analysis (PCA)
- C) Singular Value Decomposition (SVD)
- D) Backpropagation

31. Which function in PyMC is commonly used to define regression coefficients as random variables?

- A) pm.Normal()

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- B) `pm.Data()`
- C) `pm.Sampling()`
- D) `pm.ReLU()`

32. Which PyMC function is used to make predictions from a trained Bayesian regression model?

- A) `pm.sample_posterior_predictive()`
- B) `pm.traceplot()`
- C) `pm.gradient_descent()`
- D) `pm.train()`

33. Which of the following is a major challenge when applying multiple linear regression?

- A) Overfitting due to too many predictors
- B) The inability to handle categorical variables
- C) The requirement of deep learning models
- D) The inability to model linear relationships

34. What is multicollinearity in the context of multiple linear regression?

- A) When independent variables are highly correlated with each other
- B) When the dependent variable has too many missing values
- C) When regression coefficients are not statistically significant
- D) When the error term is non-Gaussian

35. How does Bayesian regression using PyMC help mitigate overfitting compared to Non-Bayesian regression?

- A) By using priors that regularize the regression coefficients
- B) By ignoring the dependent variable during inference
- C) By setting all coefficients to zero
- D) By increasing the number of independent variables

36. Which of the following is a potential drawback of Bayesian multiple linear regression in PyMC?

- A) High computational cost due to MCMC sampling
- B) Inability to estimate uncertainty
- C) Requirement for an extremely large dataset

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D) No way to incorporate prior knowledge

37. Which of the following is a common method to identify confounding variables?

- A) Checking whether an omitted variable changes the estimated effect of an independent variable significantly
- B) Looking for variables that have no correlation with the dependent variable
- C) Removing all correlated variables from the model
- D) Selecting only variables with the highest p-values

38. How can you control for confounding variables in a multiple linear regression model?

- A) By including the confounding variable as an additional independent variable
- B) By removing all independent variables except one
- C) By ignoring the confounding variable in the dataset
- D) By normalizing all variables before regression

39. What is an example of a confounding variable in a study on the relationship between exercise and weight loss?

- A) Diet, because it affects both exercise habits and weight loss
- B) Running speed, since it is an independent variable
- C) Gym membership status, since it does not directly influence the outcome
- D) The number of steps per day, since it only affects weight loss

40. What is the main reason why confounding variables can mislead causal inference?

- A) They can create an illusion of causation when there is only correlation
- B) They always lead to a model with poor predictive performance
- C) They help validate the results of a regression model
- D) They make regression coefficients more reliable

41. How does the masking effect influence multiple linear regression?

- A) It can cause some variables to appear insignificant when they actually contribute to the model
- B) It helps identify the most important features in the dataset
- C) It ensures that the dependent variable is always correctly predicted
- D) It makes all independent variables have equal influence on the outcome

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Consider the following Table for next 4 MCQs (42, 43, 44, 45)

	rank	elpd_loo	p_loo	elpd_diff	weight	se	dse	warning	scale
model_0	0	0.150551	0.001196	0.000000	0.199582	0.016608	0.000000	True	deviance
model_2	1	0.158948	0.001033	0.008397	0.198762	0.019688	0.026338	True	deviance
model_1	2	0.226300	0.003315	0.075749	0.192166	0.056111	0.056934	True	deviance
model_3	3	0.737405	0.040761	0.586854	0.149421	0.157121	0.154965	False	deviance
model_5	4	0.752202	0.032429	0.601651	0.147970	0.133582	0.138629	False	deviance
model_4	5	1.311607	0.037725	1.161056	0.112100	0.148957	0.142115	True	deviance

42. Which of the following best explains why model 4 is ranked lowest despite having the highest elpd loo value?

- A) Because a higher elpd loo always indicates worse performance
- B) Because the elpd loo values are incorrectly scaled
- C) Because ranking minimizes elpd loo, and model 4 has the highest elpd loo
- D) Because elpd loo values are on deviance scale

43. What is the most appropriate interpretation of model weights in LOO model comparison?

- A) They represent the probability that the model is true
- B) Weights are based solely on model complexity
- C) Weights sum to 1 and reflect the relative support from the data
- D) Models with warnings should be discarded regardless of weight

44. Given that model 3 and model 5 have higher elpd loo values but also higher p loo and se, what is the best inference?

- A) They are better because their predictive accuracy is higher
- B) Their higher complexity may indicate overfitting and instability
- C) They are clearly superior to model 0 and model 2
- D) The presence of no warnings proves these are the most reliable models

45. The difference in elpd loo between model 0 and model 1 is approximately 0.0757 with a standard error of 0.0561. What is the most appropriate interpretation?

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- A) Model 1 significantly outperforms model 0 in predictive accuracy
- B) Model 0 is statistically better than model 1 due to its lower elpd loo
- C) The difference in predictive accuracy between model 0 and model 1 is not statistically significant because the standard error is relatively large compared to the difference
- D) The model with higher p loo is always preferred

Solution

1. B) Simple Linear model
2. B) Multiple Logistic Regression
3. C) Softmax Regression
4. C) Polynomial Regression
5. C) Hierarchical Linear Model
6. C) Masking effect
7. C) Multicollinearity
8. D) Poisson Regression
9. B) Zero-Inflated Poisson (ZIP) Model
10. B) Hierarchical Linear Model
11. C) Include the confounding variables in the model to adjust their effects
12. B) Use uncorrelated predictors
13. C) Generalized linear model
14. B) Hierarchical Model
15. A) ψ is the chance of extra zeros, and θ is the average count of events
16. B) To assess model fit by comparing observed and predicted data
17. B) KDE (Kernel Density Estimate) plots of observed vs. predicted data with p-values
18. B) Widely Applicable Information Criterion
19. B) Predictive accuracy metrics
20. B) Leave-One-Out
21. A) Minimized for better models
22. A) Better model predictive performance
23. B) To quantify evidence in favor of one model over another
24. B) Marginal likelihoods of competing models
25. A) model 1 is better
26. B) Sensitivity to the choice of prior distributions
27. A) To model the relationship between one dependent variable and multiple independent variables
28. A) It treats model parameters as random variables with probability distributions
29. A) Normal
30. A) Markov Chain Monte Carlo (MCMC)
31. A) `pm.Normal()`
32. A) `pm.sample_posterior_predictive()`
33. A) Overfitting due to too many predictors
34. A) When independent variables are highly correlated with each other
35. A) By using priors that regularize the regression coefficients
36. A) High computational cost due to MCMC sampling

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- 37. A) Checking whether an omitted variable changes the estimated effect of an independent variable significantly
- 38. A) By including the confounding variable as an additional independent variable
- 39. A) Diet, because it affects both exercise habits and weight loss
- 40. A) They can create an illusion of causation when there is only correlation
- 41. A) It can cause some variables to appear insignificant when they actually contribute to the model
- 42. C) Because ranking minimizes elpd loo, and model 4 has the highest elpd loo
- 43. C) Weights sum to 1 and reflect the relative support from the data
- 44. B) Their higher complexity may indicate overfitting and instability
- 45. C) The difference in predictive accuracy between model 0 and model 1 is not statistically significant because the standard error is relatively large compared to the difference